

Number & Math Projects

1. **Even or Odd Checker**: Check if a given number is even or odd.
2. **Prime Number Checker**: Determine if a number is a prime number.
3. **Factorial Calculator**: Find the factorial of a number ($n!$).
4. **Fibonacci Sequence Generator**: Generate the Fibonacci sequence up to a certain number.
5. **Greatest Common Divisor (GCD)**: Find the GCD of two numbers.
6. **Least Common Multiple (LCM)**: Find the LCM of two numbers.
7. **Simple Interest Calculator**: Calculate simple interest given principal, rate, and time.
8. **Compound Interest Calculator**: Calculate compound interest.
9. **Armstrong Number Checker**: Check if a number is an Armstrong number (e.g., $153 = 1^3 + 5^3 + 3^3$).
10. **Number Reverser**: Reverse the digits of a given integer.
11. **Sum of Digits**: Calculate the sum of the digits of a number.
12. **Next Prime Number**: Find the next prime number after a given number.
13. **Binary to Decimal Converter**: Convert a binary number to its decimal equivalent.
14. **Decimal to Binary Converter**: Convert a decimal number to its binary equivalent.
15. **Area of a Circle**: Calculate the area given the radius.
16. **Collatz Conjecture Sequence**: Generate the Collatz sequence for a given number.
17. **Simple Calculator**: Create a command-line calculator that can add, subtract, multiply, and divide.
18. **Multiplication Table Generator**: Generate the multiplication table for a given number.
19. **Leap Year Checker**: Check if a given year is a leap year.
20. **Perfect Number Checker**: Check if a number is a perfect number (sum of its proper divisors equals the number itself).

Text & String Manipulation

21. **Palindrome Checker**: Check if a string is a palindrome (reads the same forwards and backward).
22. **Vowel and Consonant Counter**: Count the number of vowels and consonants in a string.
23. **Word Counter**: Count the number of words in a given sentence.
24. **Character Frequency Counter**: Count the frequency of each character in a string.
25. **String Reverser**: Reverse a given string.
26. **Case Converter**: Convert a string to uppercase or lowercase.
27. **Remove Vowels**: Remove all vowels from a string.
28. **Anagram Checker**: Check if two strings are anagrams of each other.
29. **Sort Words in a String**: Sort the words in a sentence alphabetically.

- 30. **Pig Latin Translator**: Translate a sentence into Pig Latin.
 - 31. **Acronym Generator**: Create an acronym from a phrase (e.g., "Random Access Memory" -> "RAM").
 - 32. **Email Slicer**: Extract the username and domain from an email address.
 - 33. **Count Substring Occurrences**: Count how many times a substring appears in a string.
 - 34. **Text Validator**: Check if a string contains only letters or only numbers.
 - 35. **Random Sentence Generator**: Generate a random sentence from predefined lists of nouns, verbs, and adjectives.
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🎮 Simple Games & Logic Puzzles

- 36. **Guess the Number Game**: The computer picks a number, and the user has to guess it.
 - 37. **Rock, Paper, Scissors**: Play the classic game against the computer.
 - 38. **Dice Rolling Simulator**: Simulate rolling one or more dice.
 - 39. **Coin Flip Simulator**: Simulate flipping a coin a certain number of times and show the results.
 - 40. **Hangman Game**: A simple text-based version of the Hangman game.
 - 41. **Tic-Tac-Toe (CLI)**: A command-line interface version of Tic-Tac-Toe for two players.
 - 42. **Mad Libs Generator**: Create a Mad Libs game where the user provides words to fill in the blanks.
 - 43. **Simple Quiz Game**: Create a quiz with multiple-choice questions and score the user.
 - 44. **FizzBuzz**: The classic programming challenge.
 - 45. **Number Guessing (Computer Guesses)**: You think of a number, and the computer tries to guess it using binary search.
 - 46. **Mastermind (Numbers)**: A simple version of the Mastermind game using numbers instead of colors.
 - 47. **Text Adventure Game**: A very simple story-based game where the user makes choices.
 - 48. **Higher or Lower Game**: The computer shows a number, and the user has to guess if the next one will be higher or lower.
 - 49. **Jumbled Word Game**: The program jumbles a word, and the user has to guess the correct word.
 - 50. **Blackjack (Simple)**: A basic version of Blackjack where the user plays against the dealer.
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📁 File & Data Handling

- 51. **Read a Text File**: Open and print the contents of a `.txt` file.
- 52. **Count Lines in a File**: Count the number of lines in a text file.
- 53. **Count Words in a File**: Count the number of words in a text file.
- 54. **Write to a File**: Prompt the user for input and write it to a new file.
- 55. **Append to a File**: Add new text to the end of an existing file.
- 56. **Simple To-Do List**: Create a program that lets you add, view, and delete tasks in a text file.

- 57. ****File Copier****: Copy the contents of one file to another.
 - 58. ****Find and Replace****: Find a specific word in a file and replace it with another.
 - 59. ****CSV Reader****: Read data from a `.csv` file and display it in a structured way.
 - 60. ****CSV Writer****: Write a list of data (e.g., a list of lists) to a `.csv` file.
 - 61. ****Log File Creator****: Create a simple logging function that writes messages with timestamps to a file.
 - 62. ****Merge two files****: Combine the content of two text files into one.
 - 63. ****Directory Content Lister****: List all files and directories in the current path.
 - 64. ****File Properties Viewer****: Show properties of a file (size, creation date, etc.).
 - 65. ****Configuration File Parser (Simple)****: Read key-value pairs from a simple `config.txt` file.
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🛠️ Utilities & Miscellaneous

- 66. ****Password Generator****: Generate a random password of a specified length.
 - 67. ****QR Code Generator****: Create a QR code from a given text or URL.
 - 68. ****Age Calculator****: Calculate a person's age from their date of birth.
 - 69. ****Countdown Timer****: A timer that counts down from a specified time.
 - 70. ****Simple Stopwatch****: A stopwatch that can be started, stopped, and reset.
 - 71. ****BMI Calculator****: Calculate Body Mass Index.
 - 72. ****Temperature Converter****: Convert between Celsius, Fahrenheit, and Kelvin.
 - 73. ****Currency Converter****: A simple converter using a fixed, hardcoded exchange rate.
 - 74. ****URL Shortener****: Use an API (like tinyurl) to shorten a given URL.
 - 75. ****Weather App (CLI)****: Use a free weather API to get and display the weather for a city.
 - 76. ****Internet Speed Test****: Use a library like `speedtest-cli` to check your internet speed.
 - 77. ****Alarm Clock****: A simple script that plays a sound at a specified time.
 - 78. ****Random Quote Generator****: Display a random quote from a predefined list.
 - 79. ****Plagiarism Checker (Simple)****: Compare two text files and calculate a simple similarity score.
 - 80. ****Folder Organizer****: A script that organizes files in a directory into subfolders based on their extension (e.g., `.jpg`, `.pdf`).
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📊 Beginner Data Structures

- 81. ****Find Max/Min in a List****: Find the largest and smallest numbers in a list without using `max()` or `min()`.
- 82. ****Remove Duplicates from a List****: Create a new list that contains only the unique elements of an original list.
- 83. ****List Sorter****: Sort a list of numbers without using the built-in `sort()` method (e.g., using bubble sort).
- 84. ****Common Elements Finder****: Find the common elements between two lists.

85. ****Contact Book (Dictionary)****: Create a simple contact book using a dictionary to store names and phone numbers.
86. ****Student Grade Calculator****: Use a list of dictionaries to store student names and grades, then calculate their average.
87. ****Invert a Dictionary****: Swap the keys and values of a dictionary.
88. ****Simple Shopping Cart****: Use a dictionary to represent a shopping cart where keys are items and values are quantities.
89. ****Matrix Addition****: Add two matrices (represented as lists of lists).
90. ****Flatten a Nested List****: Convert a list of lists into a single list.
91. ****Mode Finder****: Find the mode (the most frequent element) in a list.
92. ****List Comprehension Practice****: Convert several `for` loops that create lists into one-line list comprehensions.
93. ****Dictionary from two lists****: Create a dictionary from two lists, one for keys and one for values.
94. ****Check if a key exists in a dictionary****: A simple function to check for a key's presence.
95. ****Sum all items in a dictionary****: Sum up all the values in a dictionary.
96. ****Second Largest Number****: Find the second largest number in a list.
97. ****Rotate a List****: Rotate the elements of a list by a specified number of positions.
98. ****Tuple Unpacking****: Create a tuple and then unpack its values into separate variables.
99. ****Set Operations****: Demonstrate union, intersection, and difference between two sets.
100. ****Simple Stack Implementation****: Implement a stack using a list with `append()` for push and `pop()` for pop operations.